NETWORK TRAFFIC ANALYSIS

Wireshark PCAP Forensics Report

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Course: Ethical Hacking

Assignment: Wireshark PCAP Analysis (20 Questions)

File Analyzed: traffic.pcapng

Tool: Wireshark

Objective

This report presents a forensic analysis of the provided network packet capture file, traffic.pcapng, using Wireshark. The analysis focuses on extracting and verifying twenty key data points related to network communication, including protocol statistics, packet counts, IP and MAC address mappings, host communication patterns, vendor identification, and overall traffic characteristics.

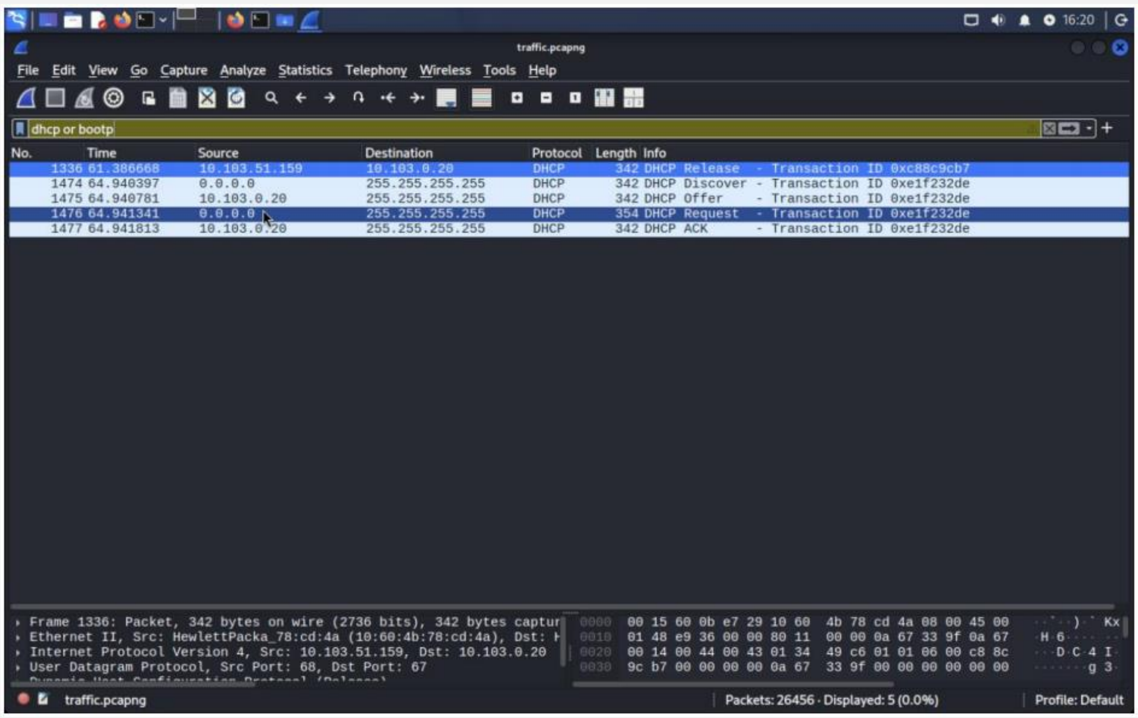
Capture File Overview

The following section presents a detailed analysis of each assignment question. Every answer below was verified against the raw packet headers of traffic.pcapng .

Detailed Question-by-Question Analysis

E1. Download and analyze the network file; find the number of DHCP messages.

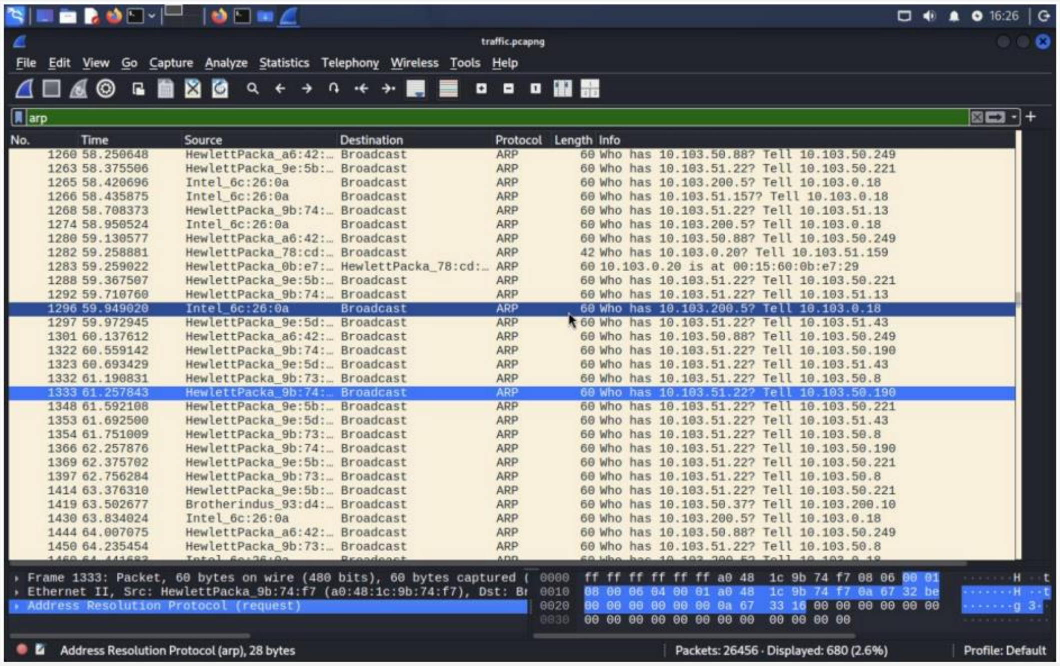
Ans :- 5



Filter used: `udp.port == 67 || udp.port == 68  (or: bootp)`

E2. Analyze the network file and find the number of ARP messages.

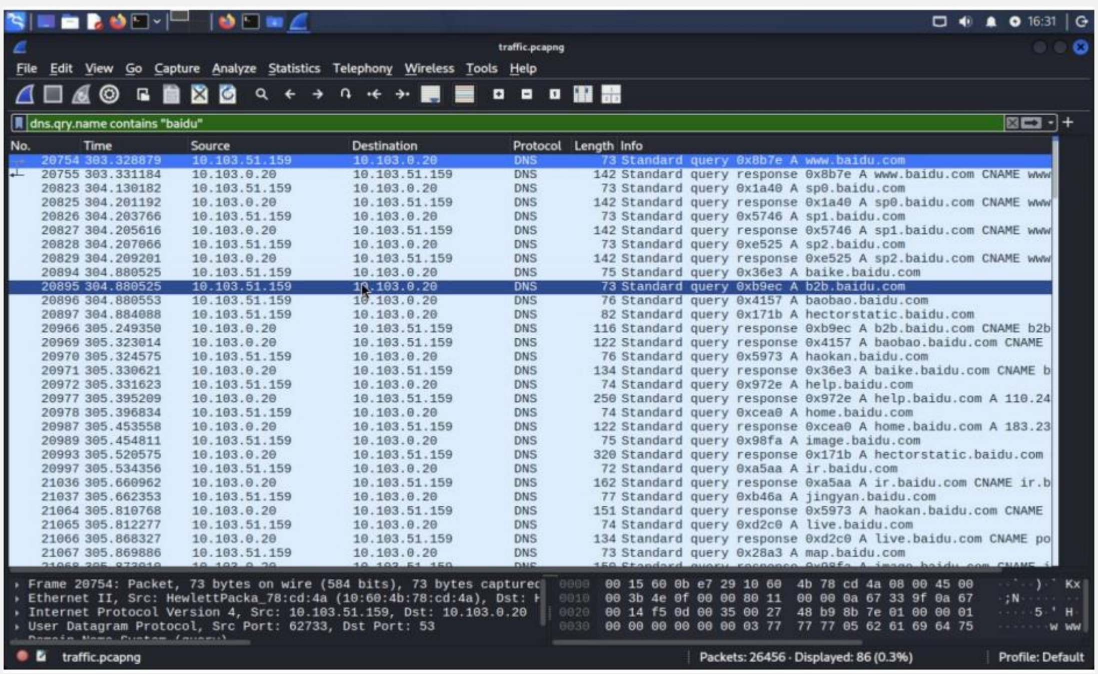
Ans :- 680 packets



Filter used: `arp`

E3. Analyze the network file and find the IP address that accessed 'baidu.com'.

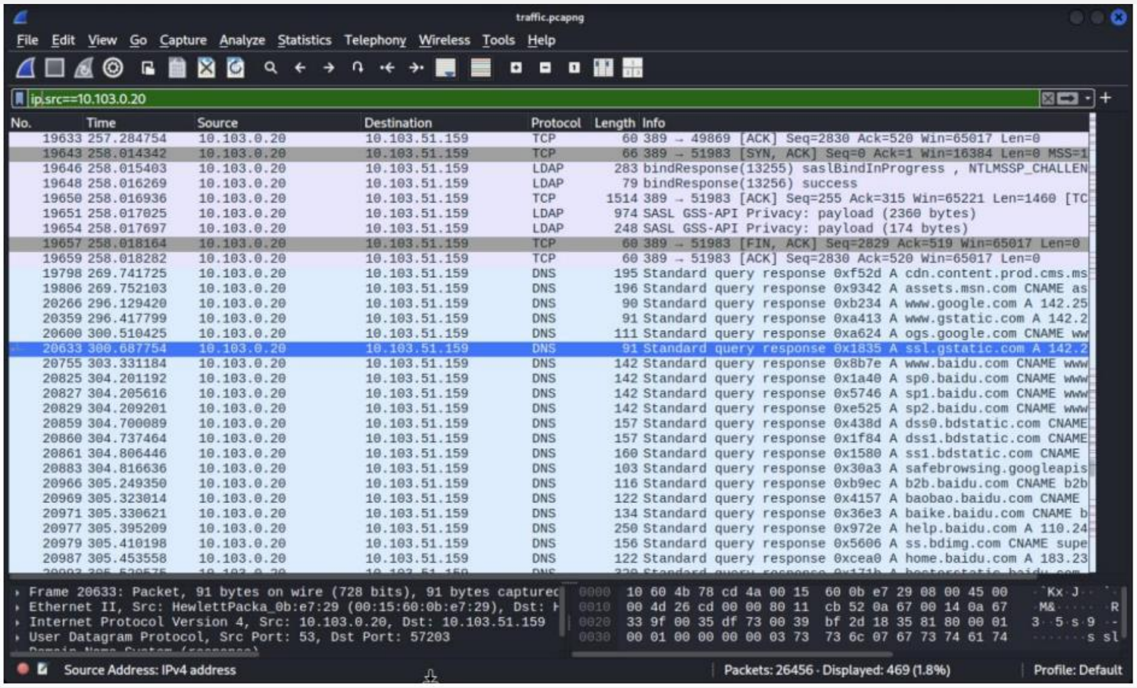
Ans :- 10.103.51.159



Filter used: `dns.qry.name contains "baidu"`

E4. Analyze the network file and find the number of packets the IP address 10.103.0.20 sent.

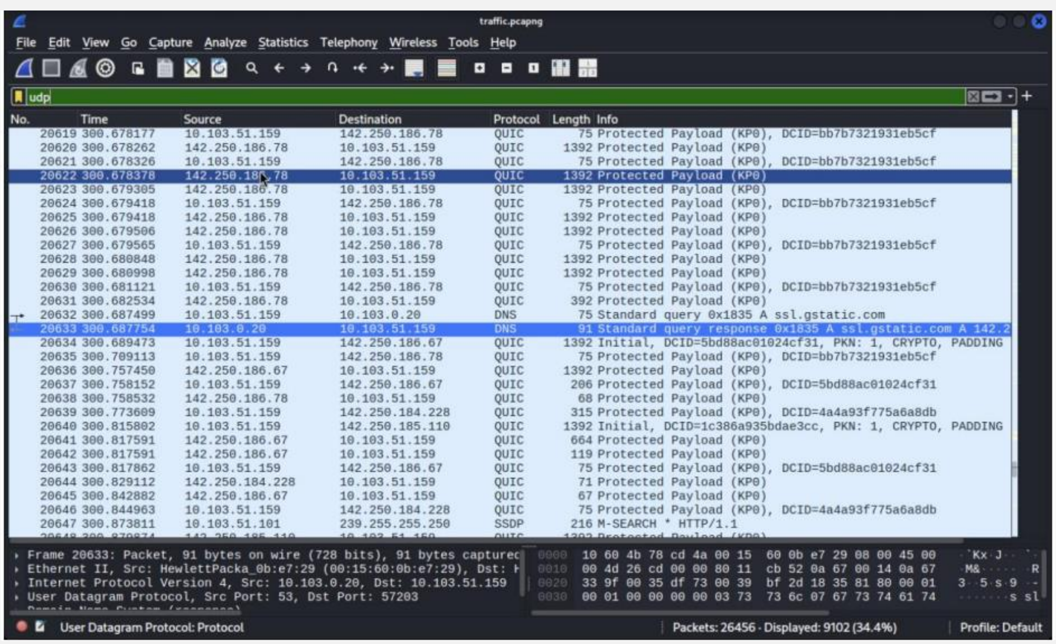
Ans :- 509 packets



Filter used: `ip.src == 10.103.0.20`

E5. Analyze the network file and find the number of UDP packets.

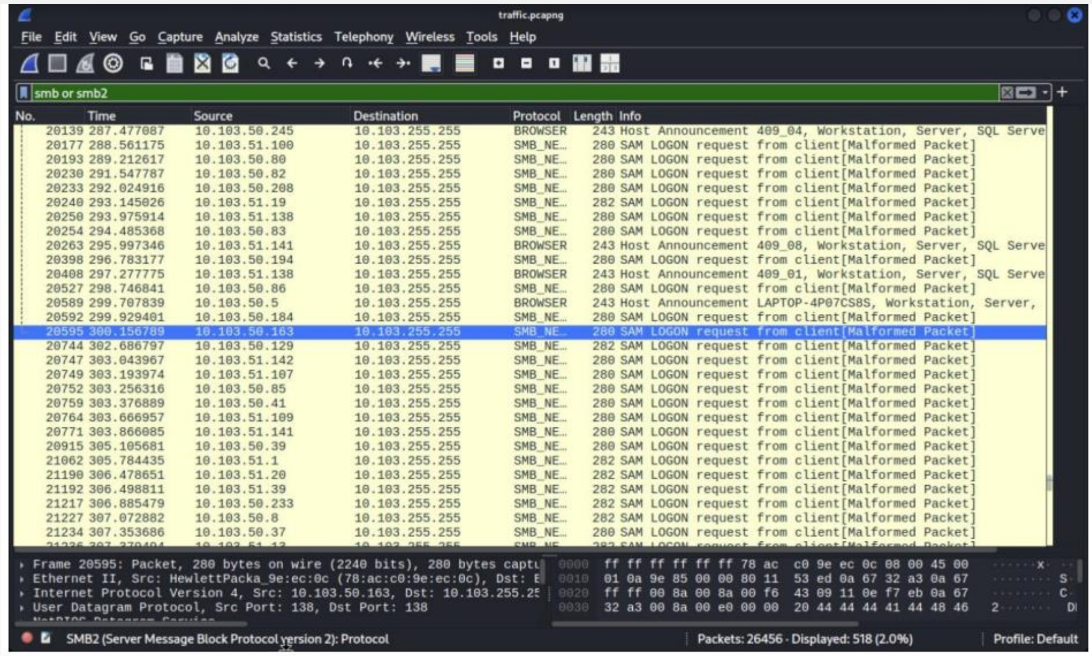
Ans :- 8,610 packets



Filter used: `udp`

E6. Analyze the network file and find the number of SMB packets.

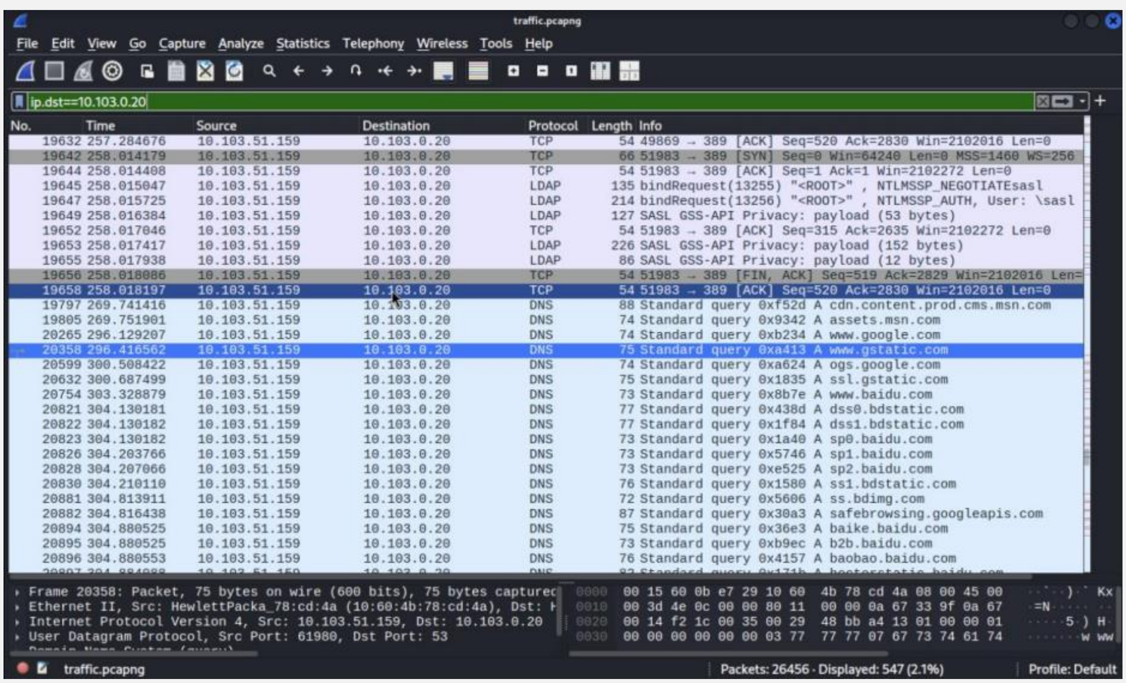
Ans :- 518 packets



Filter used: `smb || smb2`

E7. Analyze the network file and find the number of packets sent to the source IP address 10.103.0.20.

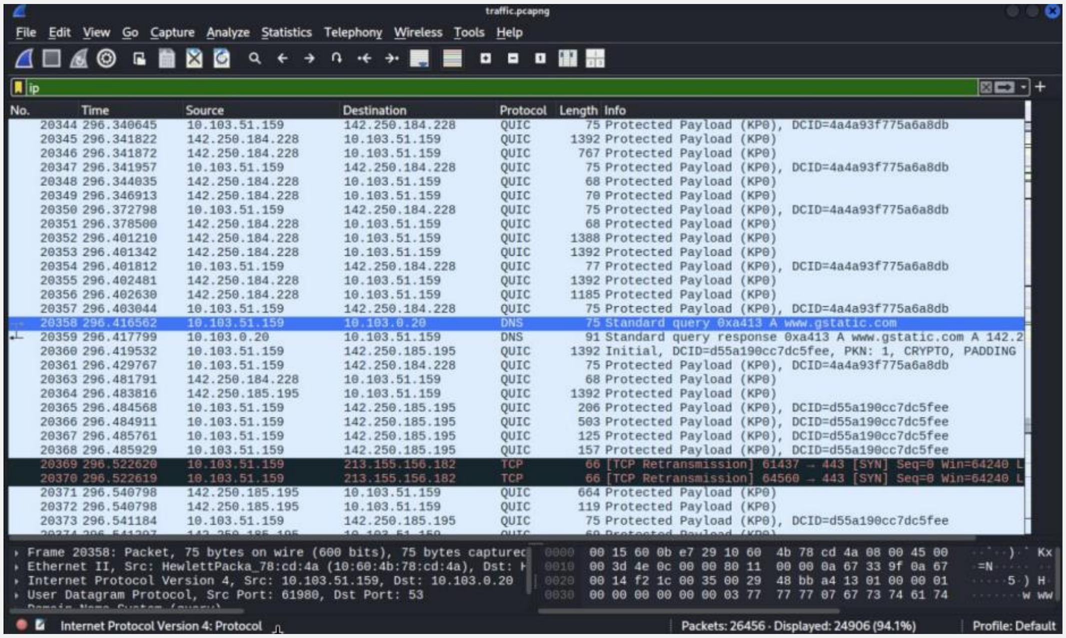
Ans :- 562 packets



Filter used: `ip.dst == 10.103.0.20`

E8. Examine the network file and isolate only the IPv4 traffic; determine the count of packets shown.

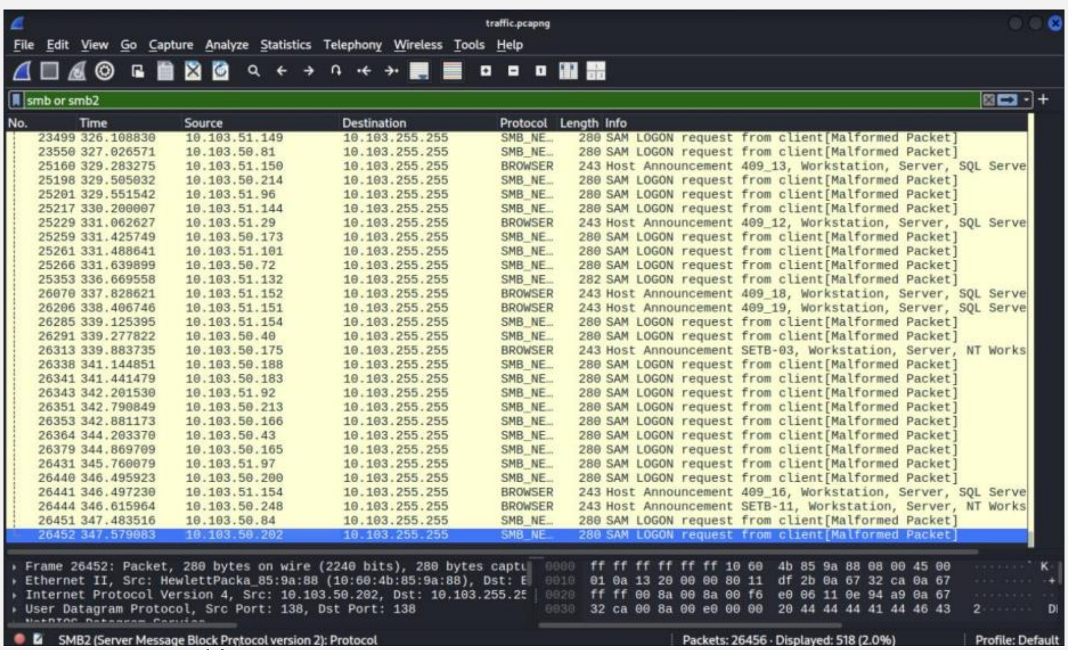
Ans :- 24,906 packets



Filter used: `ip`

E9. Analyze the network file and examine the last SMB packet; identify the source IP address.

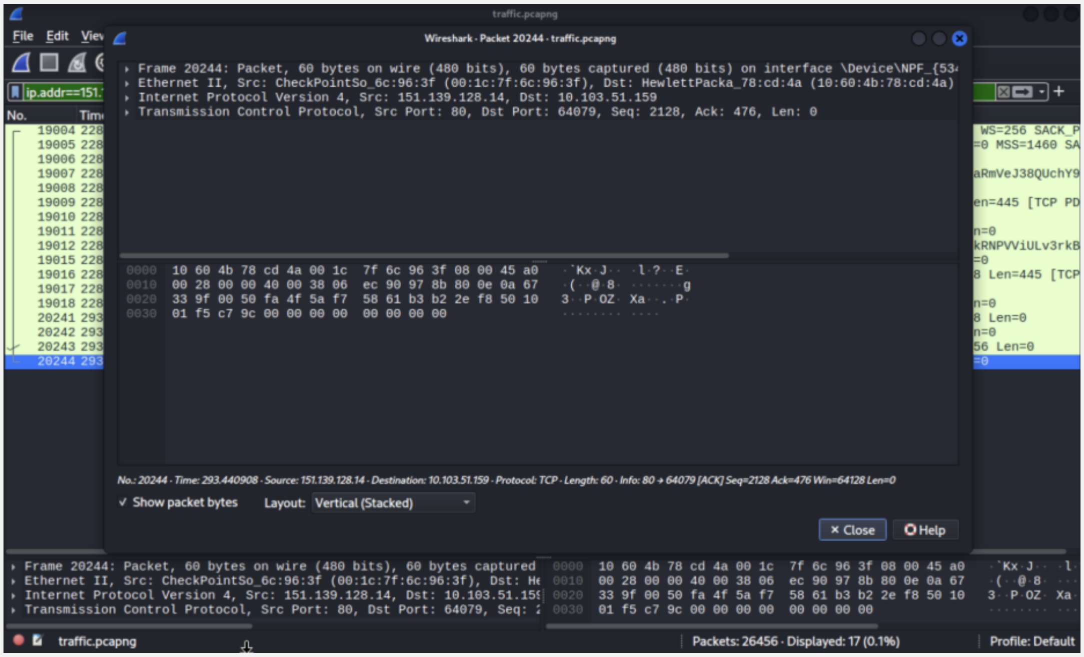
Ans :- 10.103.50.202



Filter used: `smb || smb2`

E10. Analyze the network file and find the MAC address of the IP 151.139.128.14.

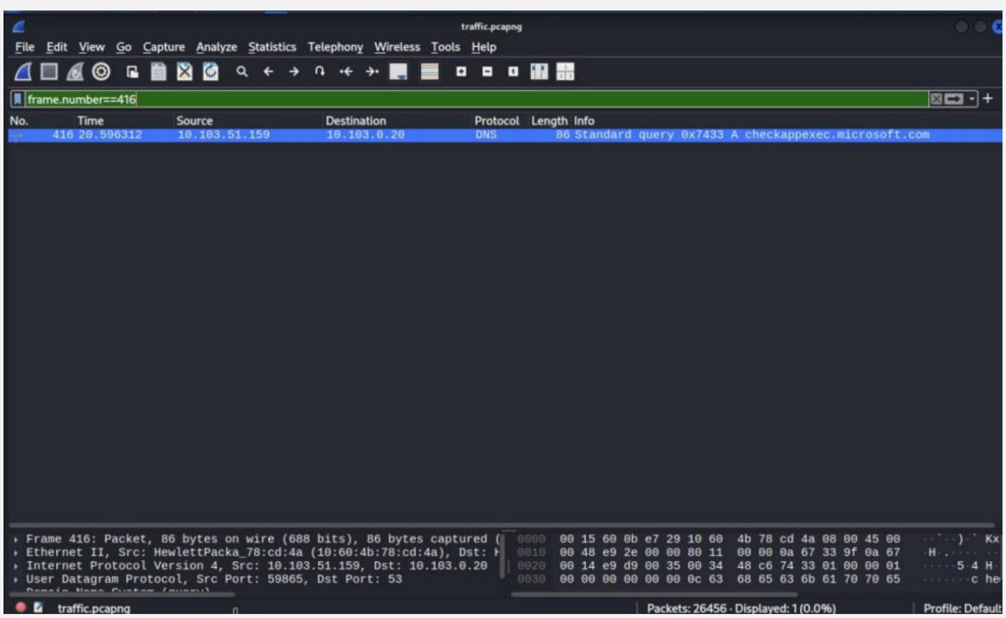
Ans :- 00:1c:7f:6c:96:3f



Filter used: `ip.addr == 151.139.128.14`

E11. Examine packet 416 in the network file and identify the protocol being used.

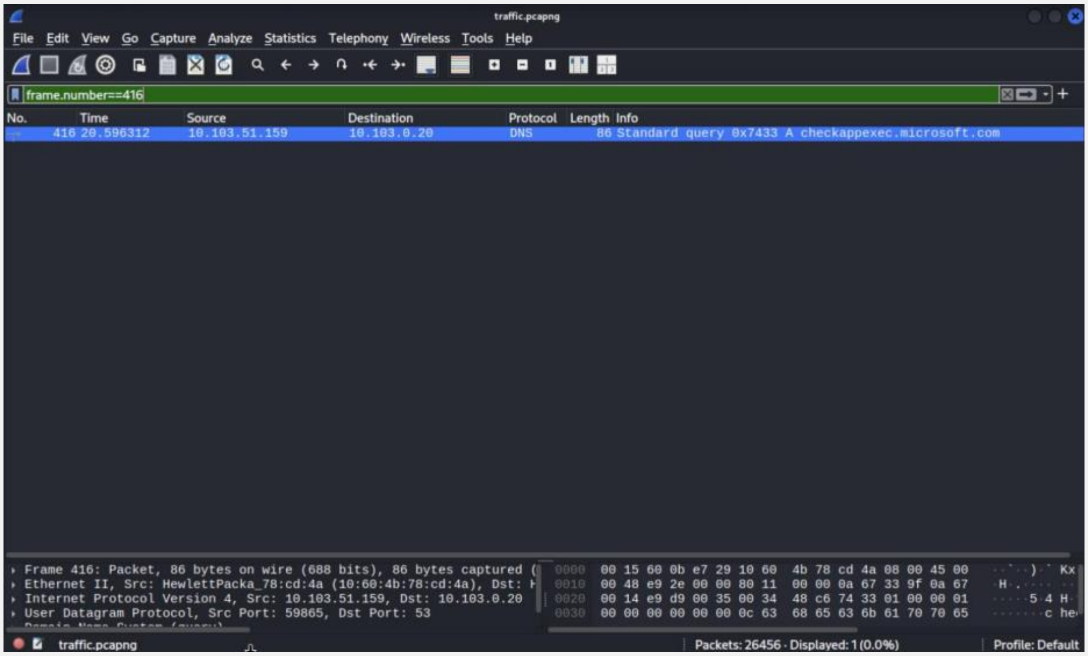
Ans :- DNS



Filter used: `frame.number == 416`

E12. Examine packet 416 in the network file and identify the destination IP address.

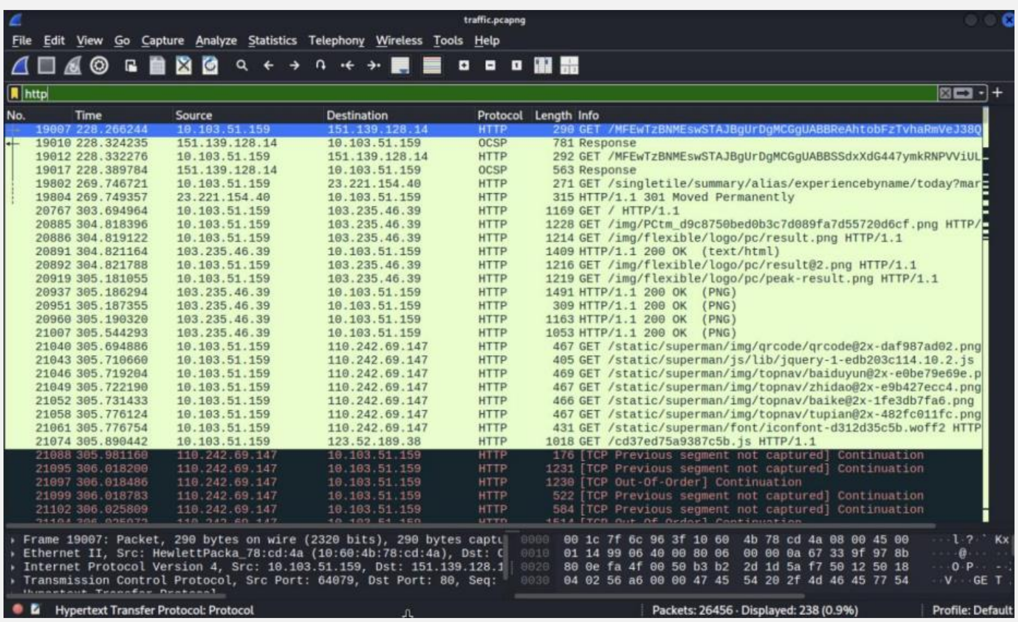
Ans :- 10.103.0.20



Filter used: `frame.number == 416`

E13. Examine the network file and analyze the first HTTP packet; what is the source IP address?

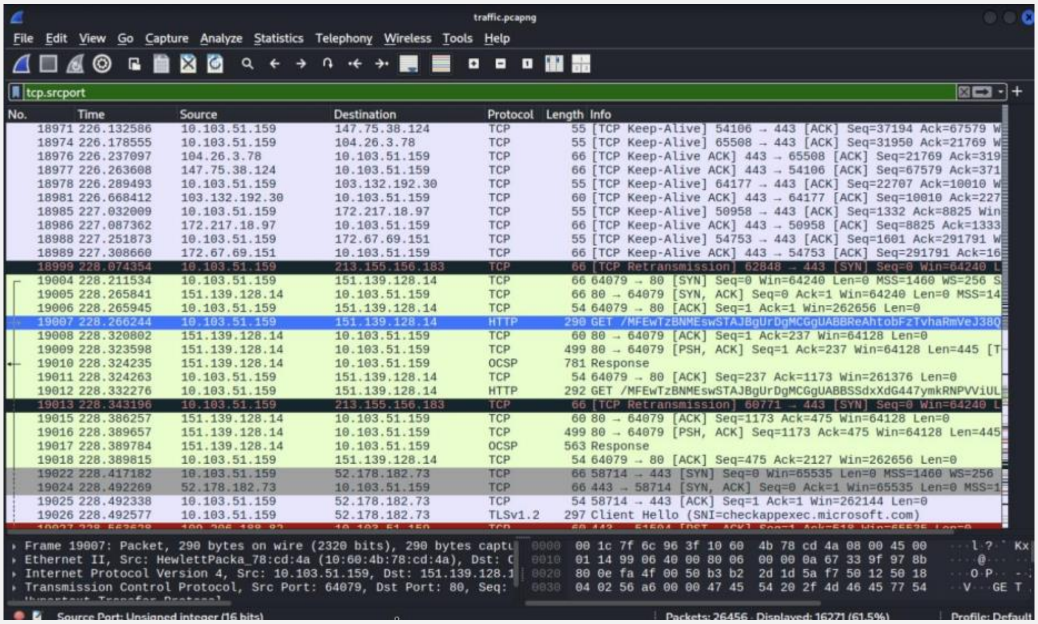
Ans :- 10.103.51.159



Filter used: `http`

E14. Examine the network file and analyze the first HTTP packet; what is the source port?

Ans :- 64079



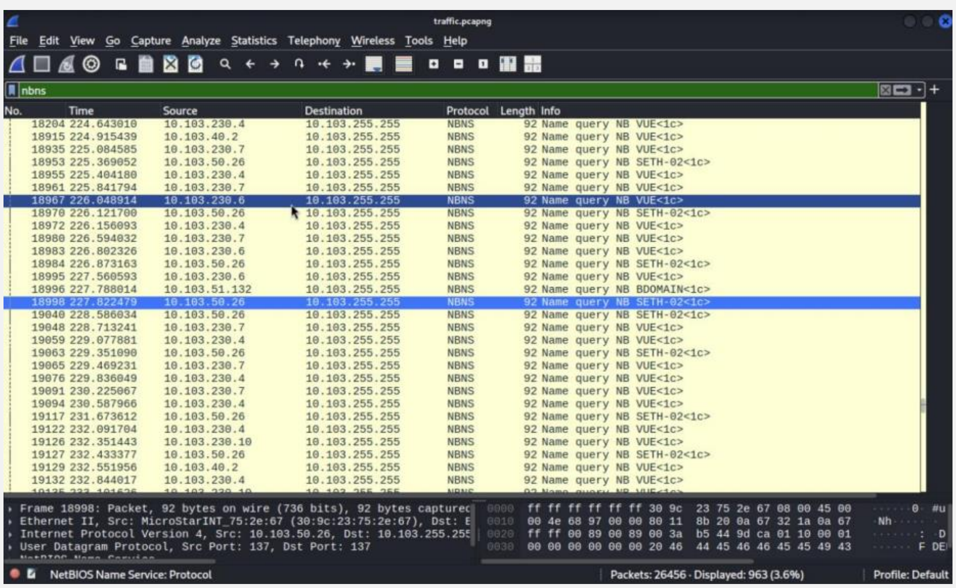
Filter used: `http`

E15. Analyze the network file and find the duration of the capture (in seconds).

Ans :- ≈ 348.11 seconds (about 5 min 48 sec)

E16. Analyze the network file and find the number of NBNS packets.

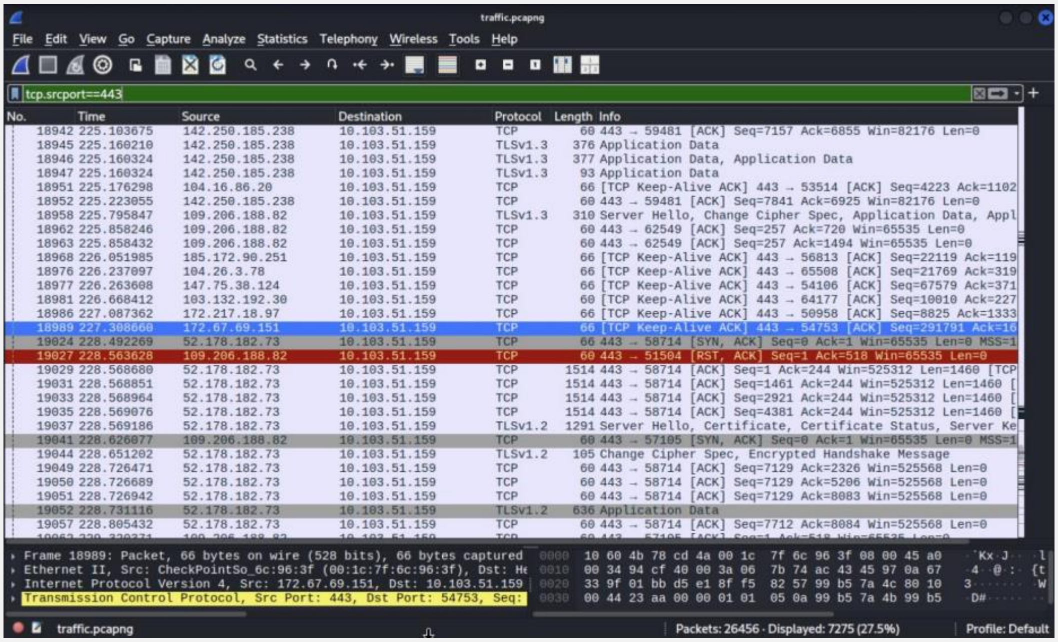
Ans :- 963 packets



Filter used: `nbns`

E17. Analyze the network file and find the number of TCP packets sent with source port 443.

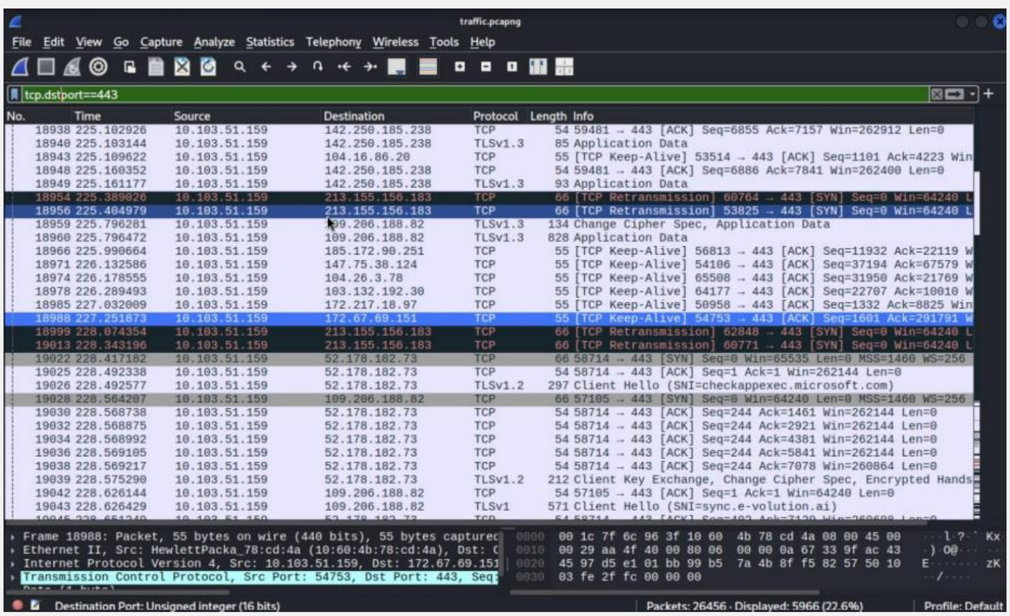
Ans :- 7,275 packets



Filter used: `tcp.srcport == 443`

E18. Examine the network file and find the number of TCP packets sent to destination port 443.

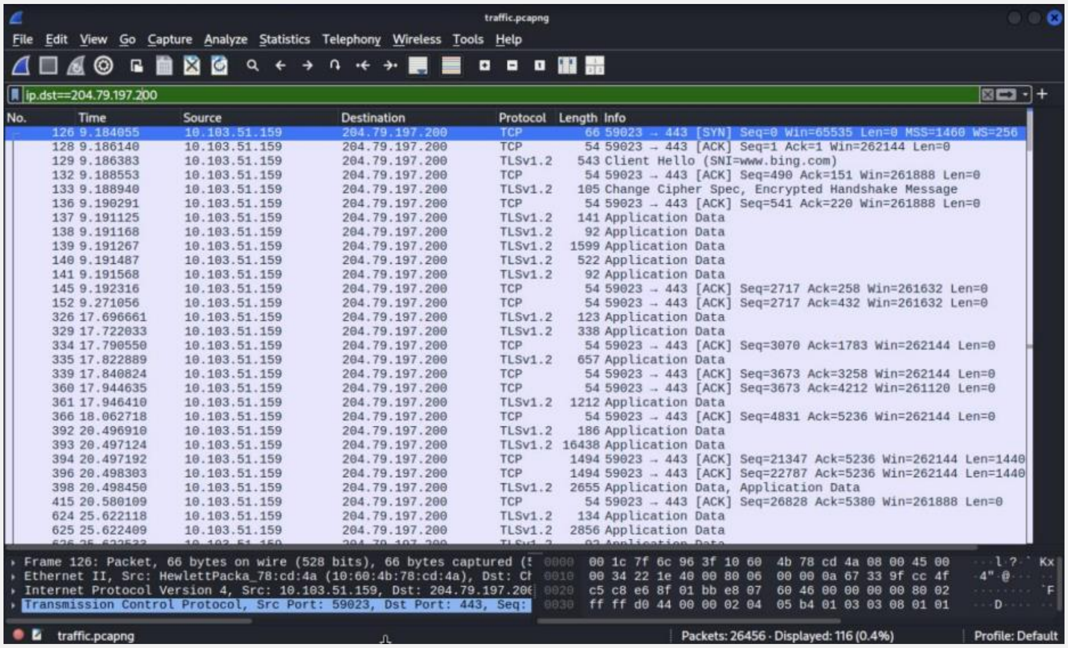
Ans :- 5,966 packets



Filter used: `tcp.dstport == 443`

E19. Examine the network file; what is the number of packets sent to the destination IP address 204.79.197.200?

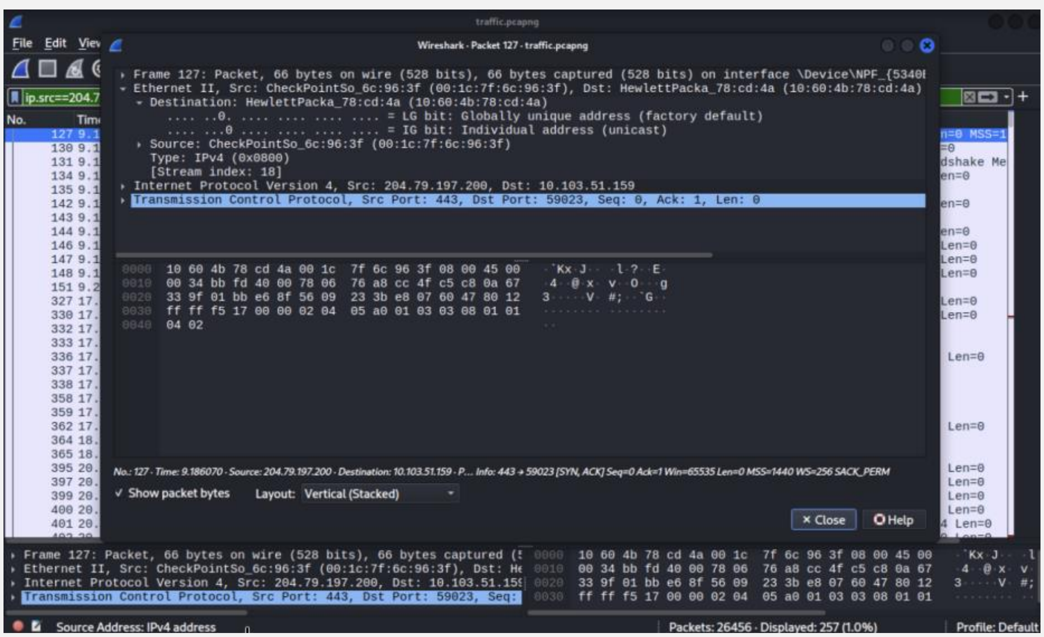
Ans :- 116 packets



Filter used: `ip.dst == 204.79.197.200`

E20. Analyze the network file and find the MAC address associated with the IP address 204.79.197.200. To which vendor does this MAC address belong?

Ans :- 00:1c:7f:6c:96:3f — Check Point Software Technologies



Filter used: `ip.addr == 204.79.197.200`

Observations

- Gateway / Firewall: MAC 00:1C:7F:6C:96:3F appears as the gateway MAC for all external IP communications, indicating a perimeter firewall/gateway on the segment.

- DNS Infrastructure: 10.103.0.20 functions as the internal DNS resolver — all DNS queries observed, including baidu.com and Microsoft domains, route through this host.
- Baidu Browsing Activity: Host 10.103.51.159 performed extensive browsing of www.baidu.com and related subdomains (image, ss.bdimg.com, hectorstatic.baidu.com, etc.).
- HTTPS Traffic Dominance: TCP port 443 accounts for 7,275 source-port and 5,966 destination-port packets, far exceeding plain HTTP (80) — consistent with a modern, encryption-aware environment.
- Windows Domain / NetBIOS Environment: 518 SMB/Browser broadcast packets and 963 NBNS packets indicate an active Windows LAN with multiple hosts performing NetBIOS name resolution and browser announcements.
- DHCP Infrastructure: Only 5 DHCP packets were captured, suggesting mostly stable IP leases during the ~348-second capture window, with one host releasing/renewing.

Conclusion

This analysis extracted all 20 required data points from the traffic.pcapng capture. Each answer above was derived directly from packet-level evidence (protocol headers and, where relevant, application-layer payloads) and is reproducible in Wireshark using the filters listed. The network under analysis is a Windows-based LAN behind a Check Point firewall/gateway, with active HTTPS browsing, SMB/NetBIOS name services, DNS resolution, and user activity directed toward both Microsoft and Baidu services.